

Multiple Choice

1. **Answer:** A

Options: A) 11.2 liters; B) 67.2 liters; C) 168 liters; D) 100.8 liters; E) 22.4 liters

Note: The correct answer is 11.2 liters because the reaction of sodium cyanide (NaCN) with dilute sulfuric acid (H_2SO_4) produces hydrogen cyanide (HCN) gas in a 1:1 mole ratio, and 196 g of NaCN corresponds to 4 moles, which at standard temperature and pressure (STP) yields a volume of 11.2 liters (4 moles x 22.4 L/mole).

2. **Answer:** B

Options: A) 0.159 pm; B) 79.5 pm; C) 210 pm; D) 132 pm; E) 0.53 pm

Note: The correct answer of 79.5 pm arises from the calculation of the expectation value of the electron's position in the ground state of the hydrogen atom, which is determined by the quantum mechanical model and the wavefunction associated with the electron's orbital.

3. **Answer:** C

Options: A) 1000 H^+ ions; B) 900 H^+ ions; C) 843 H^+ ions; D) 950 H^+ ions; E) 615 H^+ ions

Note: The correct answer of 843 H^+ ions is derived from calculating the volume of the mitochondrion as a sphere.

3. **Answer:** A

Options: A) It contains 20 protons and neutrons; B) All atoms of calcium have 20 protons and 40 neutrons; C) It contains 20 protons, neutrons, and electrons; D) All calcium atoms have 20 protons, neutrons, and electrons; E) All atoms of calcium have a mass of 40.078 u

Note: The correct answer is based on the fact that a calcium atom has an atomic number of 20, indicating it contains 20 protons, and its common isotope, calcium-40, has a total of 40 nucleons, which includes both protons and neutrons, confirming the presence of 20 neutrons.

4. **Answer:** C

Options: A) H-O; B) P-Cl; C) N-H; D) S-O; E) C-O

Note: The N-H bond is not polar because nitrogen is more electronegative than hydrogen, leading to nitrogen being written first, while in polar bonds, the more electronegative atom is typically written last.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because electronegativity generally increases across a

period and decreases down a group, and although silver is located further to the right in the periodic table than iron, it is actually positioned below it, leading to a lower electronegativity value for silver (1.93) compared to iron (1.83).

7. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the dimensions of the tank, when converted to meters (2.0 m, 0.5 m, and 0.2 m), multiply to give a volume of 0.2 m³, which is equivalent to 200 liters, not 250 liters.

8. **Answer:** True

Options: A) True; B) False

Note: CBr₄ has a tetrahedral molecular geometry because the carbon atom forms four equivalent single bonds with bromine atoms, resulting in a balanced distribution of electron density that minimizes repulsion according to VSEPR theory.

9. **Answer:** False

Options: A) True; B) False

Note: The conjugate acid of the H₂PO₄⁻ ion is H₃PO₄, not H₄PO₄⁻, because a conjugate acid is formed by adding a proton (H⁺) to the base, which in this case results in H₃PO₄ rather than H₄PO₄⁻.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the frequency of light can be calculated using the equation $f = \frac{c}{\lambda}$, where c is the speed of light (approximately 3.00×10^8 m/s) and λ is the wavelength (632.8 nm or 632.8×10^{-9} m), resulting in a frequency of approximately 5.750×10^{14} Hz.

Long Form Response

11. **Answer:** To neutralize 28.0 g of hydrochloric acid (HCl), we first determine the moles of HCl using its molar mass (36.46 g/mol), yielding 0.767 moles. The balanced reaction with calcium hydroxide (Ca(OH)) is: $2 \text{HCl} + \text{Ca(OH)} \rightarrow \text{CaCl} + 2 \text{HO}$, indicating a 2:1 stoichiometric ratio. Therefore, 0.767 moles of HCl requires 0.384 moles of Ca(OH), which corresponds to 28.4 g of Ca(OH) (molar mass 74.09 g/mol). Next, considering the reaction between Ca(OH) and phosphoric acid (HPO), which follows the equation: $3 \text{Ca(OH)} + 2 \text{HPO} \rightarrow \text{Ca(PO)} + 6 \text{HO}$, we find that 0.384 moles of Ca(OH) can neutralize 0.256 moles of HPO. This results in a mass of 25.1 g of HPO (molar mass 98.00 g/mol) that can be neutralized, demonstrating the application of stoichiometric principles in acid-base reactions.

Note: The answer correctly applies stoichiometric principles to determine the moles of hydrochloric acid and the corresponding mass of calcium hydroxide required for neutralization, while also recognizing the balanced chemical reactions that dictate the stoichiometric ratios involved in both neutralization processes.

12. **Answer:** In a binary mixture of water and ethanol, the chemical potentials of the components are interdependent, governed by Raoult's Law and the Gibbs-Duhem relation. At a mole fraction of water (x_{water}) equal to 0.60, if the chemical potential of water changes, it affects the chemical potential of ethanol due to the conservation of total chemical potential in the system. Specifically, a change in water's chemical potential of 0.15 J mol^{-1} results in a corresponding adjustment in ethanol's chemical potential, as the two must equilibrate. *Note: correct because the interdependence of chemical potentials in a binary mixture, as described by Raoult's Law and the Gibbs-Duhem relation, ensures that any change in the chemical potential of one component, such as water, directly affects the chemical potential of the other component, ethanol.*

End of Answer Key